A supersingular auxiliary curve for PKM's isogeny tower

research/pkm_supersingular_aux_20261015 · 2026-10-15 · branch `claude/pkm-supersingular-aux`
Protocol `PROTOCOL.md` frozen in `8d719fddf` (sha256 `db109df9…c1ad65`) before any measured cell

1. Question and verdicts

Over the NIST P-192 prime ($p \equiv 3 \bmod 4$, $v_2(p+1) = 64$) the supersingular curve $y^2 = x^3 + x$ has $p+1$ points and a cyclic 2-Sylow subgroup, so a point of order 2^{48} costs one draw, against about 11.2 core-years for the ordinary curve the backlog card `XP-20261015-p192-isogeny-aux-search-t48` prices. Does PKM's isogeny-tower factor base work as well on it? Three pre-registered questions:

question	verdict	class
Q1: is $V_{\text{ss}} = x(R' + \langle Q \rangle)$ the Dickson torus set up to a Möbius map?	**No; by the rule's letter, partial.** At $t = 3…5$ over 103 primes, 0 of 6,507 sets of $y^2 = x^3 + x$, 0 of 6,768 $j = 0$ sets and 0 of 24,580 ordinary sets match a torus set; 2 of 12,969 sets of $y^2 = x^3 - x$ do, at one cell ($p = 383$, $t = 3$). At $t = 2$ every family matches about 75 % (one modulus).	exact
Q2: F4 solving degree D, supersingular vs ordinary vs torus	**H_same fails; supersingular = ordinary.** All four supersingular curves give the ordinary isogeny tower's D in every cell ($m = 2$, $N \le 20$; $m = 3, 4$, $N \le 12$; three primes); the torus differs (4 against 5 at $m = 2$, $N = 4…10$; 5 against 6 at $m = 3$, $N = 6$).	measured
Q3: relation phase, charged (ecbench)	**The search cost goes, nothing else changes.** Aux share 0.9–1.9 % of S (build only) against 10–27 % for the searched ordinary curve; S within 2.2 % of a plain base of the same size; yield unchanged. ss ÷ ordinary 1.11 / 0.82 / 0.74 on s20 / s22 / s24 (s20: larger base).	measured

For the card (derived). Nothing measured distinguishes a supersingular E' from an ordinary one except the cost of finding it, so the ordinary search is not needed to feed the relation phase. The gate of any P-192 use, lead 1 (bounded D as N grows), is untouched; nothing here concerns P-192's security.

2. Q1: the structure of V (exact)

For every prime $p \equiv 3 \bmod 4$ in $[100, 3000]$ and $t = 2…5$ with $t + 1 \le v_2(p+1)$, every tower set of every family (all Dickson torus cosets; every $x(R' + \langle Q \rangle)$ on $y^2 = x^3 \pm x$, and on $y^2 = x^3 + 1$, $y^2 = x^3 + d^3$ when $p \equiv 2 \bmod 3$; three ordinary curves; the positive control $y^2 = x^3 + 16x \cong y^2 = x^3 + x$) is reduced to a complete $\mathrm{PGL}_2(\mathbb{F}_p)$ invariant and compared.

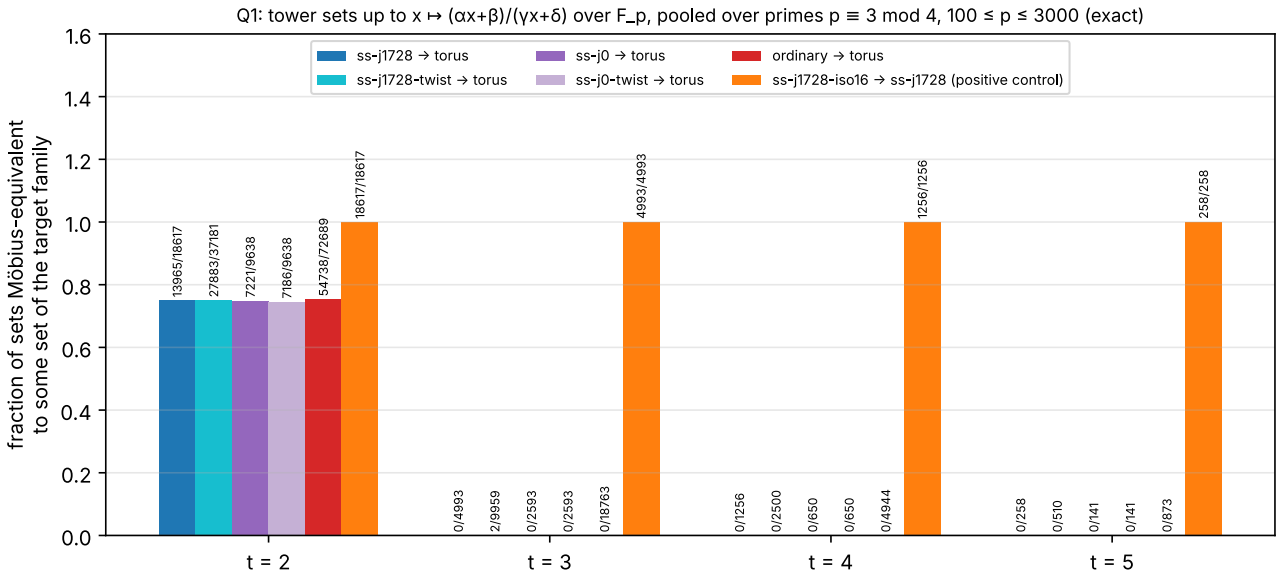


Figure 1: Fraction of each family's sets equivalent to some torus set (positive control: to some $y^2 = x^3 + x$ set), pooled over primes; labels are counts. Source `q1/pooled.tsv`.

t	cells	$x^3 + x$	$x^3 - x$	$j = 0$ (both)	ordinary	pos. control
2	103	13965 / 18617	27883 / 37181	14407 / 19276	54738 / 72689	18617 / 18617
3	53	0 / 4993	**2** / 9959	0 / 5186	0 / 18763	4993 / 4993
4	27	0 / 1256	0 / 2500	0 / 1300	0 / 4944	1256 / 1256
5	12	0 / 258	0 / 510	0 / 282	0 / 873	258 / 258

Sets equivalent to a torus set / all sets (positive control: to a $y^2 = x^3 + x$ set). The canonical form agreed with brute force over $\mathrm{PGL}_2(\mathbb{F}_p)$ on all 906 checked pairs ($p \le 131$). The one exception at $t \ge 3$ is a single shared class at $p = 383$; the pre-registered rule therefore reads **partial**, but no family-wide map exists on the data, and none for the P-192 curve $y^2 = x^3 + x$.

Why. The torus set is a fibre of the polynomial D_{2t}, totally ramified over ∞; the supersingular set is a fibre of a composed Vélu x-map, whose ramification indices are at most 2. The maps are not Möbius-conjugate; single fibres can still match by coincidence, which happens only at $t = 2$ (one cross-ratio modulus) and for every pair of families alike.

3. Q2: F4 solving degree (measured)

`f4_fp_tower` (reduced presentation, full-rank exit, degree bound $n + d + 6$), two random targets per cell, one thread; 336 systems, none stopped; 296 tower rows ($m \le 3$) agree with an exhaustive recount over V^m. Primes $T_{24} = 7 \cdot 2^{21} - 1$, $T_{31} = 61 \cdot 2^{25} - 1$, $S_{25} = 9 \cdot 2^{21} - 1$, and the positive control at $p_0 = 3 \cdot 2^{18} + 1$.

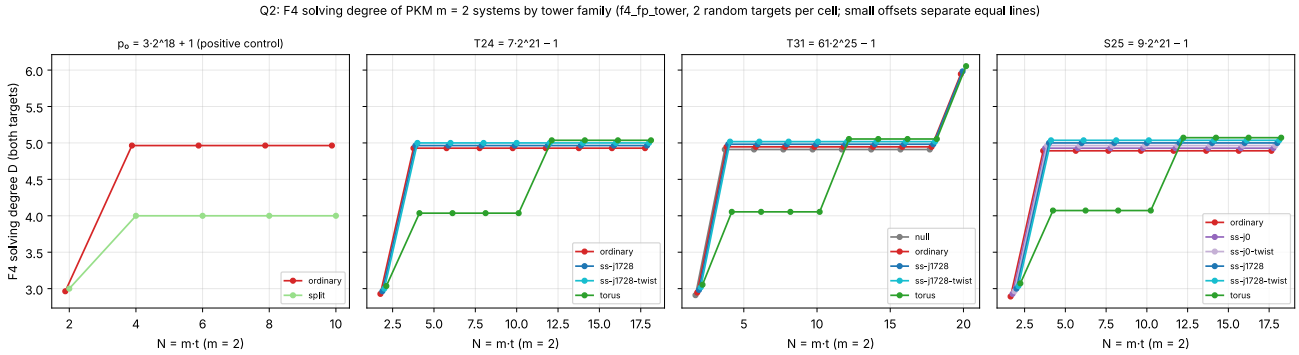


Figure 2: D against N at $m = 2$; both targets equal in every cell. Source `q2/cells.tsv`.

N	ss-j1728	ss twists, j=0	ordinary	torus	null on ss	p_0: ordinary / split
2	3/4/2	3/4/2	3/4/2	3/4/2	3/4/2	3 / 3

4	5/16/4	5/16/4	5/16/4	**4** /16/4	5/16/4	5 / 4
6	5/57/7	5/57/7	5/57/7	**4** /56/6	5/57/7	5 / 4
8	5/150/9	5/150/9	5/150/9	**4** /152/8	5/150/9	5 / 4
10	5/346/12	5/346/12	5/346/12	**4** /336/10	5/346/11	5 / 4
12	5/1182/16	5/1181/16	5/1181/16	5/1223/13	5/1182/15	
14	5/2629/19	5/2629/19	5/2629/19	5/2303/15	5/2628/18	
16	5/5034/22	5/5034/22	5/5034/22	5/4786/17	5/5034/21	
18	5/9220/24	5/9221/24	5/9221/24	5/8639/20	5/9184/23	
20	6/35157/28	—	6/35158/28	6/33426/24	—	

Entries D / width / steps ($m = 2$), identical at every prime where the arm ran. At $m = 3$: ss 6, 6, 6 and ordinary 6, 6, 6 against torus 5, 6, 6 ($N = 6, 9, 12$); at $m = 4$: all 6, 7 ($N = 8, 12$). The torus rows reproduce round 8 (at $N = 20$ its multiply-add count to the unit). **Why:** every Vélu tower has top forms $y_j y_{j+1} - y_j^2$; Dickson towers have $-y_j^2$.

4. Q3: relation phase, charged (measured)

Three P-192-shape curves ($v_2(p-1) = 1$, $v_2(p+1) = 10$, 20–24 bits), one ecbench session each, 7 arms × 8 workloads × 2 measured rounds; 504/504 runs verified, every replay identical, L0 (operation counts only).

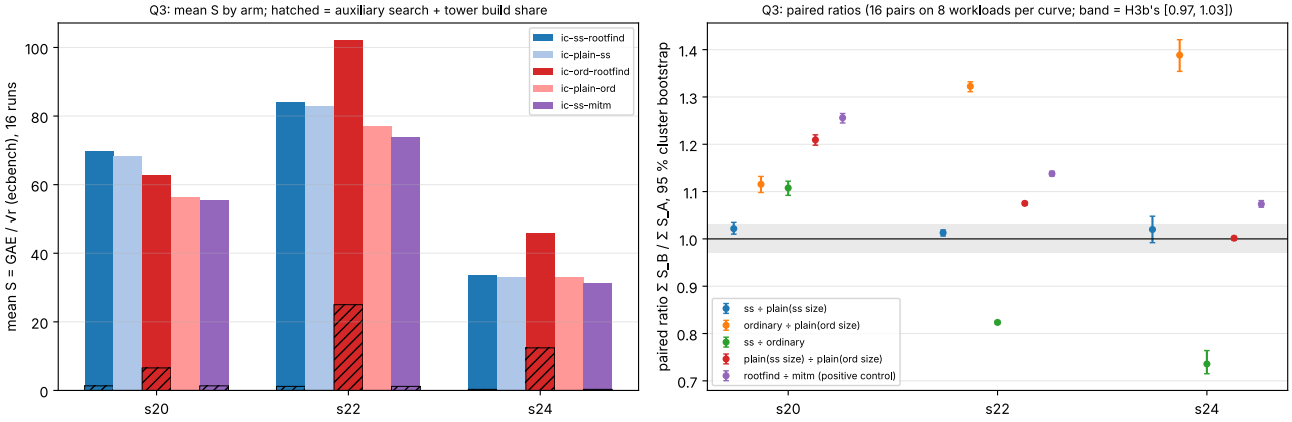


Figure 3: Left: mean S per arm (16 runs), hatched = auxiliary search + build. Right: paired ratios with ecbench's 95 % cluster-bootstrap intervals. Source `q3/tables/`.

curve	ss ÷ plain(ss)	ord ÷ plain(ord)	ss ÷ ord	plain(ss) ÷ plain(ord)	aux share ss / ord	rootfind ÷ mitm
s20	1.022 [1.010, 1.035]	1.115 [1.098, 1.132]	1.108 [1.092, 1.122]	1.209	1.9 % / 10.4 %	1.256
s22	1.013 [1.006, 1.019]	1.322 [1.311, 1.332]	0.824 [0.820, 0.827]	1.075	1.4 % / 24.5 %	1.138
s24	1.020 [0.992, 1.048]	1.388 [1.354, 1.421]	0.736 [0.715, 0.764]	1.002	0.9 % / 27.1 %	1.074

Rules: P1 held; H3a held; H3b held on s22, s24 and failed on s20 (2.2 % excess, the build's 1.9 % share); H3c held on s22, s24 and failed on s20 (base 141 against 123 abscissae; size-corrected 0.92); H3d held (yield within 1.8 %); controls passed (A/A 1.000; rootfind ÷ mitm above 1). Every IC arm is 29–94× rho with the negation map.

5. Scope

Small fields only ($p < 2^{31}$ for F4, $p < 2^{24}$ for ecbench); $N \leq 20$; one seed per arm in Q3. Q1 is exact on the primes run, not a proof for all p. The registry gained three curves; its derived artifacts are not regenerated in this change. Two container restarts are recorded in `progress.txt`.